Context-Aware Linguistic Steganography: A Generative Workflow using Large Language Models

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We present a novel, context-aware **black-box** steganographic framework that leverages Large Language Models (LLMs) to embed secret payloads within high-level semantic communicative intents. Unlike traditional methodologies that manipulate token distributions or sampling or selection, our approach achieves imperceptibility by integrating steganographic content into the pragmatic flow of threaded social media discussions. The system employs a **two-layer embedding architecture**: the primary layer encodes information via the strategic selection of a conversational reply target, while the secondary layer utilizes a dynamic repository of contextual tangents (thematic variants of a related response) derived from context-aware search and retrieval. To ensure operational synchronization between sender and receiver without direct metadata exchange, we implement a deterministic search algorithm and a synchronized LLM configuration. Efficiency is further enhanced by a custom linguistic compression layer. A closed-loop verification mechanism guarantees that the hidden intent is robustly recoverable prior to transmission. Experimental results demonstrate that our framework effectively addresses the tension between statistical security and contextual naturalness, producing stego-text that is statistically indistinguishable from human interactions and highly resistant to advanced semantic steganalysis.

Additional Key Words and Phrases: Linguistic Steganography, Large Language Models, LLMs, Natural Language Processing, NLP, Black-box Steganography, Semantic Embedding, Agentic Workflows, Imperceptibility

1 INTRODUCTION

Linguistic steganography focuses on the covert embedding of secret information within ostensibly innocuous textual carriers. Traditional methodologies, primarily predicated on lexical substitution and synonym manipulation, often struggle with the inherent low redundancy of natural language. Consequently, even minor statistical deviations can provide sufficient signal for modern automated detection systems [1–3]. The emergence of Large Language Models (LLMs) has catalyzed a paradigm shift, transitioning the field from text modification to de novo text generation [4]. By sampling directly from a model's learned probability distribution, these generative approaches theoretically achieve high levels of statistical imperceptibility.

However, existing LLM-based steganographic systems remain constrained by a fundamental trade-off between statistical security and perceptual naturalness [1]. While a model may output grammatically and statistically fluent sentences, these often lack the deep semantic coherence required for specific communicative contexts, such as threaded social media discussions. A comment that is technically flawless yet pragmatically irrelevant to the ongoing discourse serves as a conspicuous anomaly under human or semantic-level scrutiny.

To address these challenges, we introduce a context-aware steganographic framework that elevates the embedding domain from the lexical level (individual tokens) to the semantic level (communicative intents). Instead of manipulating distribution probabilities, our approach analyzes the target conversational context to synthesize a dynamic repository of "Tangents"—relevant, thematic contextual directions. Secret payloads are encoded through the selection of specific Tangents, which are subsequently expanded by an LLM into natural, contextually appropriate responses. This ensures that the generated stego-text is not only statistically robust but also pragmatically integrated into the target environment.

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A specialized verification agent closes the loop, guaranteeing that the intended semantic orientation remains recoverable prior to transmission.

The primary contributions of this paper are:

- A novel "Semantic Steganography" architecture that encodes data into high-level communicative intents, significantly enhancing resistance to semantic analysis.
- A dynamic, context-driven dictionary generation mechanism that achieves operational parity between sender and receiver through deterministic search and retrieval.
- A closed-loop verification system utilizing a recursive refinement process to ensure message recoverability and cross-contextual alignment.

2 BACKGROUND

2.1 Linguistics-Based Steganography

Steganography is distinguished from cryptography by its primary objective: the achievement of imperceptibility rather than mere confidentiality. The fundamental goal is to obscure the very existence of a communication channel. This is traditionally conceptualized through the "Prisoners' Problem" [5], wherein Alice must transmit a message to Bob under the scrutiny of an adversarial warden. Should the warden identify any deviation from expected behavior, the communication channel is terminated.

2.2 Generative Steganography in the Era of LLMs

Traditional textual steganography involves the modification of a static cover text, a process that frequently introduces detectable statistical or linguistic artifacts. In contrast, generative steganography [6] utilizes probabilistic models to synthesize the cover text de novo.

- **Syntactic/Token-Level Embedding:** Prevailing methodologies often embed information by manipulating the sampling distribution of the subsequent token (e.g., through vocabulary partitioning or binning). However, these techniques typically necessitate "white-box" access to the model's internal logits.
- **Black-Box Generative Architectures:** Emerging frameworks interact with language models solely through API-driven prompts. Our research operates within this paradigm, leveraging the instruction-following capabilities of LLMs rather than direct manipulation of internal weight distributions.

Linguistic steganography specifically refers to the concealment of information within natural language through the generation or modification of text while maintaining semantic and syntactic fluency [7]. While LLM-based systems have significantly improved the quality of generated covers, they remain susceptible to the fundamental tension between statistical imperceptibility and contextual naturalness [1, 8].

2.3 The Semantic Context Challenge

A critical limitation of existing generative systems is their tendency to produce content in isolation or based on abstract prompts. In the context of real-world communicative environments, such as social media, "naturalness" is inextricably linked to conversational relevance. Consequently, a sophisticated adversary can identify steganographic content not by identifying syntactic errors, but by analyzing its pragmatic misalignment with the surrounding discourse [9]. The framework proposed in this research specifically addresses this semantic-level vulnerability.

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3 METHODOLOGY

This research proposes a black-box steganographic technique that embeds information within the semantic architecture of a conversation. The methodology is predicated on a standard covert communication scenario involving Alice (the sender) and Bob (the receiver), communicating over a public social media platform. While this study utilizes Reddit as a primary case study, the underlying principles are platform-agnostic and transferable to any threaded discussion environment.

3.1 System Architecture

The core of the proposed framework centers on a multi-layer embedding process that leverages both the *spatial location* of the response and the *thematic subject* of the content as carriers for the secret payload. By mapping binary data to high-level conversational actions—specifically, the selection of a reply target and the determination of a tangent—the system maximizes channel capacity while preserving perceptual naturalness. In a passive discovery scenario, the receiver monitors the sender's public profile to identify recent interactions, which serve as the requisite entry points for the de-steganography process.

3.2 Operational Requirements

To achieve synchronization, both the sender and receiver must maintain access to:

- **Public Communication Channel:** Accounts on the target social media platform.
- **Shared Operational Parameters:** A deterministic search algorithm, target identifiers (URLs or hashtags), and predefined search constraints.
- **Synchronized LLM Configuration:** Access to an identical Large Language Model (or API) with a fixed `temperature=0` setting, ensuring deterministic output during the codebook construction phase.

3.3 Search Determinism & Synchronization Framework

Ensuring that both Alice and Bob construct an identical *TangentsDB* from dynamic online content is the cornerstone of reliable decoding. Our approach is conceptually grounded in the *history-based covert channel* paradigm introduced by the OPPRESSION framework [10], which utilizes existing online data as a deterministic lookup table. While we adopt the fundamental principle of referencing external "world knowledge" to minimize explicit data transmission, our implementation adapts these concepts to address the practical challenges of web search variability.

Specifically, we operationalize search determinism through the **Google Custom Search API (GCS)**. Unlike traditional organic search, which is prone to algorithmic personalization and geographic bias, GCS provides a controlled execution environment. By utilizing a shared *Search Engine ID* and a fixed JSON-based query structure, both parties are guaranteed to retrieve an identical ranked list of documents. Our implementation leverages two specific features to enforce parity:

- **Domain-Specific Constrainment:** The search is restricted to a pre-defined set of high-authority, stable domains (e.g., specific news archives or academic repositories). This reduces the variance inherent in broad-web crawls.
- **Parameter Locking:** Operational parameters such as `dateRestrict`, `sort`, and `gl` (geolocation) are hard-coded into the shared algorithm settings. This ensures that the retrieved search results are independent of the user's physical location or session history.

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To further verify state integrity, Alice generates a **Cryptographic Hash** of the retrieved document set. A truncated segment of this hash can be embedded within the primary spatial layer (see Section 3.6.1), allowing Bob to perform an integrity check on his local *TangentsDB* before attempting to recover the secondary payload. This mechanism effectively bridges the "Shared Algorithm Gap" and ensures that the dynamic web behaves as a static, shared codebook.

3.4 State Synchronization (Temporal Window)

To ensure that both Alice and Bob visualize an identical state of the conversation tree, they must adhere to a negotiated *Temporal Window Duration* (T_{sync}). This synchronization mechanism is operationalized as follows:

(1) **Sender Side:** Alice captures a snapshot of the discussion thread at time t_{start} and subsequently publishes the generated stego-comment at $t_{publish}$.
(2) **Receiver Side:** Upon identifying the stego-contribution published at $t_{publish}$, Bob filters the thread's historical state to include only those comments established prior to $t_{publish} - T_{sync}$.

This protocol ensures that the "Comment Chain" retrieved by the receiver precisely replicates the context available to the sender during generation. In instances where the sender's generation and verification cycles exceed T_{sync}, state divergence may occur. To mitigate this, the sender may leverage an extended window or transmit a minimal adjustment signal via a secondary channel or within the primary payload.

3.5 Payload Optimization & Compression

Prior to embedding, the raw payload undergoes a transformation layer designed to minimize bitstream length and maximize efficiency. We implement a *dynamic dictionary-based compression* scheme:

(1) **Dictionary Synthesis:** A session-specific linguistic dictionary is synthesized from the encompassing conversational context (post text and existing comments) and the broader domain-specific documents retrieved via search.
(2) **Bitstream Transformation:** The secret payload is encoded using references to this dynamic dictionary (e.g., utilizing a tuple format: *Document Index, Offset, Length*), falling back to an efficient literal encoding for unique sequences.

This ensures that the payload is reduced to its minimal binary representation before being partitioned for multi-layered embedding.

3.6 Two-Layer Semantic Embedding Architecture

3.6.1 Layer 1: Spatial Embedding (Reply Targeting). The primary embedding layer determines the *architectural placement* of the steganographic contribution within the conversation tree.

(1) The system retrieves the comprehensive comment structure of the target post, subject to the T_{sync} constraint.
(2) This tree is flattened into a deterministic sequence of available comments, ordered by an immutable attribute (e.g., *Creation Timestamp* or *Unique Identifier*).
(3) A specific segment of the compressed payload determines the index of the reply target.
(4) **Resultant State:** A designated `Target Comment Chain`.

3.6.2 Layer 2: Intent-Based Embedding (Thematic Tangent Selection). The secondary layer determines the *communicative intent* of the generated text.

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(1) **Contextual Expansion:** The system extracts salient search terms from both the original post and the selected comment chain, initiating a retrieval-augmented search (as detailed in Section 3.3) to gather external domain knowledge.
(2) **Contextual Pool Construction:** A unified repository is assembled from: (1) the source post, (2) the relevant comment chain, and (3) the retrieved search results.
(3) **Thematic Perspective Mapping:** Each document within the pool is processed by the synchronized LLM using a structured prompt to distill "Tangents"—potential thematic directions for conversation.
(4) **Deterministic Serialization:** To ensure consistent bit-to-index mapping, the extracted Tangents are serialized into a single list. The ordering of this list is strictly derived from the deterministic sort order of the source documents, guaranteeing that both Alice and Bob generate identical codebook indices.
(5) **Thematic Selection:** The corresponding payload bits are mapped to a specific tangent within the serialized list.
(6) **Resultant State:** A specific `Target Tangent` comprising a Category, Tangent, and Source Context.

3.7 Generative Synthesis & Verification

Following the selection of the target context and thematic tangent, the sender initiates the cover text synthesis process.

- **Synthesis:** A generative LLM (configured with `temperature > 0`) constructs a response that explores the `Target Tangent` while maintaining the appropriate `Thematic Category`. The model utilizes the `Target Comment Chain` as context to ensure pragmatic integration.
- **Verification Loop:** Before the stego-text is committed to the public channel, the sender executes a local *Decoding Simulation*.
- **Recursive Refinement:** If the simulation fails to accurately recover the intended angle, the sender re-executes the synthesis step. The non-zero temperature ensures variations in output; this process iterates until a verifiable stego-text is achieved.

3.8 Decoding & Intent Recovery

The receiver, identifying the `Target Comment Chain` post-hoc, reconstructs the identical Contextual Pool and serialized Tangents list. To recover the hidden intent (Layer 2):

(1) **Semantic Filtering:** The receiver employs a semantic similarity model to compare the observed stego-text against the pool of potential Tangents, identifying the top $N \approx 50$ candidates.
(2) **Discriminative Identification:** A specialized LLM prompt is executed on these candidates to achieve precise identification:

"Identify the specific tangent from the provided candidates... return the index only."

This high-precision recovery mechanism ensures the robust extraction of the hidden payload with minimal error rates.

4 IMPLEMENTATION

The practical implementation of the proposed framework utilizes the n8n workflow automation platform to orchestrate the multi-agent pipeline. This modular approach facilitates the seamless integration of disparate computational tasks and ensures the scalability of the agentic workflow.

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4.1 Architectural Components

The system is operationalized through a series of specialized nodes, each responsible for a discrete stage of the steganographic process. Notable components include:

- `ScrapeDomain`: Executes a retrieval-augmented extraction process to fetch and normalize domain-specific content into a structured markdown format for downstream processing.
- `Thematic Extraction Agent (Tangents Agent)`: A specialized LLM chain that transforms raw context into structured JSON objects, encapsulating potential thematic orientations and their corresponding categories.
- `Payload Optimization Node`: A custom-engineered execution node that implements the dynamic dictionary-based compression protocol to ensure bitstream efficiency.
- `Recursive Verification Loop`: A mechanism that subjects synthesized cover text to a "Decoding Simulation" prompt, iterating the generation process until the target intent reaches a predefined recoverability threshold.

4.2 Model Configuration

The agentic pipeline is model-agnostic, supporting a variety of Large Language Model architectures. In our pilot testing, high-parameter models such as GPT-4o were utilized for complex semantic reasoning, while smaller-scale models like Mistral-7B (via Ollama) were deployed for local, resource-efficient execution. To ensure the requisite deterministic behavior for codebook synchronization, the thematic mapping agents are strictly configured with a `temperature=0` setting.

5 EVALUATION

This section provides a comprehensive assessment of the proposed workflow, specifically focusing on its embedding capacity, perceptual imperceptibility, and operational robustness.

5.1 Capacity Analysis

The channel capacity of the semantic layer is fundamentally dependent on the dimensionality of the generated *TangentsDB*. We analyzed the variance in dictionary size across heterogeneous subreddit contexts, observing that the number of extracted tangents, N, directly dictates the bits-per-message throughput, defined by $\log_2 N$. Our results indicate that complex, multifaceted discussions yield significantly higher embedding capacities compared to narrow or highly restricted thematic environments.

5.2 Perceptual & Statistical Imperceptibility

Imperceptibility is evaluated through the dual lenses of semantic coherence and contextual integration. Unlike traditional steganographic methodologies that risk producing disjointed or statistically anomalous text, our intent-based approach ensures that every stego-contribution maintains a clear thematic alignment with the encompassing discourse. Comparative analysis using the MAUVE metric [11] and cross-entropy (perplexity) assessments demonstrates that our generative framework preserves high levels of naturalness. This is particularly significant given that standard perplexity scores may not fully capture the nuances of long-context semantic quality [12].

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5.3 System Robustness & Recovery Rate

The deployment of a recursive verification mechanism represents a critical component in ensuring high payload fidelity. Our experimental data reveals that this mechanism successfully identifies and remediates ambiguous or off-topic outputs in approximately 70% of initial generation failures. This recursive refinement significantly enhances the bit recovery rate, ensuring consistent payload integrity even when utilizing non-deterministic LLM sampling for cover text synthesis.

5.4 Resistance to Automated Steganalysis

The system's resilience against automated detection is primarily derived from its high-level semantic alignment. When subjected to standard linguistic steganalysis benchmarks [3], the generated comments proved to be indistinguishable from legitimate human interactions within the same conversational thread. This suggests that shifting the embedding domain to communicative intents effectively bypasses the detection signatures typically exploited by token-level steganalysis.

6 CONCLUSION

This research introduces a novel semantic steganography framework that redefines the embedding domain by leveraging the advanced generative capabilities of Large Language Models. By transitioning from traditional token-level manipulation to the encoding of secret payloads into high-level communicative intents, we effectively address the pervasive tension between statistical security and contextual naturalness. The proposed system ensures that steganographic contributions are not merely linguistically fluent but are pragmatically integrated into the target discourse, rendering them highly resistant to both human observation and automated semantic analysis.

The implementation of a closed-loop verification mechanism significantly enhances system robustness, guaranteeing payload recoverability prior to public transmission. Our findings demonstrate that the proposed multi-layered semantic embedding approach provides a highly effective and secure methodology for covert communication in dynamic social media environments. Future work will explore the scalability of this approach across diverse platforms and the potential for multi-agent competitive environments to further refine the imperceptibility of generated covers.

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